



Department of Computer Science and Engineering

Academic Year 2022– 2023(Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8603 & Distributed Systems

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit 4 / Peterson–Kearns algorithm based on vector time, Helary-Mostefaoui-Netzer-Raynal Communication -induced Protocol
- **Course Outcome:** The student will be able to describe the agreement protocols and fault tolerance mechanisms in distributed systems
- **Topic Learning Outcome:** TLO10.
- **Activity Chosen:** Cross Word Puzzle
- **Justification:**
 - The synchronous checkpointing algorithm is an important concept of distributed systems. To identify the learning level of students about the Checkpointing algorithm this activity was given. It is an important topic the question was asked in most of the university questions.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - The activity was conducted in the class room with 52 number of students. The announcement was given to the students in-prior.
 - In a class room,a printed sheet of 10 questions to each student given separately to the students. The students filled the answer in the given sheet and returned back to me.
 - The answer for the wrong questions discussed in the class room immediately to make them to know answer for all questions.
 - The sample cross word puzzle activity presented by the students as shown in the figure 1 and figure 2.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO1	PSO2	PSO3
CO4	3	2	1	1	2	2	3	3	2

(1 – Low

2 – Moderate

3 – High)



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• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO10	PO12	PSO1	PSO2	PSO3
	3	2	1	1	2	2	3	3	2
Justification for correlation	The basic knowledge of engineering fundamentals is necessary to get the knowledge in Synchronizing Checkpointing algorithm	The student should identify the basic knowledge of checkpointing algorithm for solving computer engineering problems	The student will be Provide the solution for complex problem with the knowledge of checkpointing algorithm in distributed systems	The student will be able Use the knowledge of checkpointing algorithm in distributed computing	The student doing the problem by individual	The students improve communication by sharing knowledge in the form of presentation	The concept of Checkpointing algorithm is important in mobile application development	The student will be able to provide the distributed solution using the knowledge of Checkpointing algorithm	The student will be able to solve AI and big data analytics problems

- ## Screen shots





- **Reflective Critique:**

- **Feedback of practice from students and other stakeholders:**

- The students felt that this activity used to recall information.
 - The student said this activity reduces the stress and strengthen our mind.

- **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- The students a chance to evaluate their knowledge in addition to encouraging them to focus on terminology because they must spell every word correctly.
 - Interaction with their class mates during the class was effective.
 - The students able to improve analytical skills and speedup the learning capacity.
 - With the help of this activity, each student can get clear idea about the need of checkpointing algorithm in distributed systems.

- ❖ **Challenges faced in implementation:**

- Some students do not know the answer for the question number 3.
 - The activity was planned for 15 Minutes, some students not yet completed within stipulated time, it took 25 minutes.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/>
- <https://www.theshoresoflakephalen.com/benefits-of-crossword-puzzles/>
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/IG_CrosswordPuzzle.pdf

Signature of Faculty Member

HOD